

Appendix for Prompt Optimization with Verifiable Rewards for Synthetic Essay Generation

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A LLM Prompts for Synthetic Essay Generation

A.1 Templates for Synthetic Essay Generation

Your input fields are:

1. 'target_score' (UnionType[int, str]): Target 'final score' for the simulated essay based on the rubric. Note that in some essay prompts, 'final score' is a total composite score of individual writing rubric traits with smaller point allocations (e.g., 1-4 for each trait, and 5-20 for the total score from the five traits). This can be a single integer point or a score range.
2. 'essay_prompt' (str): Prompt for the simulated essay.

Your output fields are:

1. 'starting_sentences' (dict[Literal['A', 'B', 'C', 'D', 'E'], str]): 5 starting sentences for the essay.
2. 'starting_sentence_pick' (str): Randomly pick one for the starting sentences. Return a single letter, without any formatting, from A to E.
3. 'generated_essay' (str): Write the essay with the selected starting sentence, without the index letter.

All interactions will be structured in the following way, with the appropriate values filled in.

```
[[ ## target_score ## ]]  
{target_score}
```

```
[[ ## essay_prompt ## ]]  
{essay_prompt}
```

```
[[ ## starting_sentences ## ]]  
{starting_sentences}          # note: the value you produce must adhere to  
the JSON schema: {"type": "object", "additionalProperties": {"type":  
  "string"}, "propertyNames": {"enum": ["A", "B", "C", "D", "E"]}}
```

```
[[ ## starting_sentence_pick ## ]]
```

```

{starting_sentence_pick}

[[ ## generated_essay ## ]]
{generated_essay}

[[ ## completed ## ]]
In adhering to this structure, your objective is:
{Optimized LLM Prompt}

[[ ## target_score ## ]]
{target score}

[[ ## essay_prompt ## ]]
{Essay set descriptions}
{Scoring rubric}

Respond with the corresponding output fields, starting with the field ‘[[
  ## starting_sentences ## ]]' (must be formatted as a valid Python
  dict[Literal['A', 'B', 'C', 'D', 'E'], str]), then ‘[[ ##
  starting_sentence_pick ## ]]', then ‘[[ ## generated_essay ## ]]',
  and then ending with the marker for ‘[[ ## completed ## ]]'.
```

A.2 Unoptimized Prompt Instructions

This instruction was used for the Base and Feature-Guided conditions.

```

You are an expert essay tutor. Simulate the student essays for a given
score level.
The essay does not need to start with the title.
Do not begin with: The, In, I, It, This, There, As, From, Since
Do not begin with a quotation or part of the writing prompt's title.
Do not use tool in any outputs. Follow the formatting instructions.
```

A.3 Automatically Optimized Instructions

This is an example instruction optimized for Essay Set 2, using `word_count` as the optimization target, with GPT-OSS-20B as the generator.

```

**Task: Generate a 10th-grade persuasive editorial on library censorship
**

You will receive a **single JSON-style object** containing the following
keys:

| Key                               | Description |
|-----|-----|
| **target_score**         | Integer 1-6 - the rubric score the essay must
  achieve. |
```

```

| **essay_prompt**          | Full prompt text (includes quotation and
|                           | rubric). |
| **related_stats**        | For each possible score (1-6) a *mean* and *
|                           | standard deviation* for 'word_count'. |
| **starting_sentences**   | Dictionary of five candidate opening
|                           | sentences labelled "A"-"E". |
| **starting_sentence_pick** | The key (e.g., "C") that tells you which
|                           | opening sentence to use. |

Your job is to **write ONE essay** that satisfies **all** constraints
below.
Output **only** the essay text (plain paragraphs). No titles, quotations,
JSON, headings, tool tags, or any explanatory metadata may appear.

---

### 1. Opening sentence
- The essay **must begin** with the exact sentence from '
  starting_sentences' whose key equals 'starting_sentence_pick'.
- There must be **no characters** (including spaces, line-breaks,
  invisible Unicode) before this sentence.
- If that sentence starts with any **prohibited first word** (case-
  insensitive): 'The', 'In', 'I', 'It', 'This', 'There', 'As', 'From',
  'Since', prepend a **neutral bridge** of **3-4 words** (e.g., "Indeed
  ," "Consider this," "Reflect on this") that **does not** itself start
  with a prohibited word. Keep the original opening sentence unchanged
  after the bridge.

### 2. Word-count window
1. Retrieve the 'mean' and 'std' for the given 'target_score' from '
  related_stats'.
2. Compute:
  'lower = mean - std'
  'upper = mean + std'
  (Treat these as real numbers; do **not** round.)
3. The final essay's total word count **must satisfy** 'lower\u202f<=\\
  u202fcount\u202f<=\\u202fupper' (inclusive).
  - Count words by splitting the final text on **whitespace**; every
  token separated by whitespace counts as one word.

### 3. Stylistic length expectations (use as a guide **in addition** to
the statistical window)
| Score | Approx. words | Typical features |
|-----|-----|-----|
| 1 | 50-120 | Very brief, simple/repetitive sentences, minimal
organization. |
| 2 | 130-250 | Short sentences, weak transitions, limited detail. |
| 3 | 250-400 | Clear intro/body/conclusion, some supporting details,
occasional complex sentences. |

```

4	400-550	Clear thesis, logical paragraph order, varied sentences, few lapses.
5	560-700	Strong cohesion, frequent complex sentences, precise vocabulary, minor slips.
6	700-1000	Sophisticated, nuanced argument, rich vocabulary, multiple rhetorical devices, flawless organization.

If the statistical window allows a length outside the range above, ****still write**** so that the essay **reads** like a genuine piece at the required score level (adjust detail, sentence complexity, etc.).

4. Content requirements

- ****Topic:**** Persuade a newspaper readership about ****censorship in libraries****.
- ****Position:**** Clearly state ****for or against**** the removal of offensive material.
- ****Arguments:**** Provide ****at least two distinct arguments****.
 - Each argument must be illustrated with a ****personal experience, observation, or a specific reading/example****.
- ****Conclusion:**** End with a paragraph that restates your stance and offers a final thought or call-to-action.

5. Tone & voice

- Use a ****formal-personal register**** appropriate for a 10th-grade student.
- The sophistication of vocabulary, sentence structures, and rhetorical devices should match the ****target_score**** (simpler & direct for low scores, confident & nuanced for high scores).

6. Organization (recommended structure)

1. ****Intro paragraph**** - starts with the required opening sentence, immediately states your position.
2. ****Body paragraphs**** - 2-3 paragraphs, each presenting ****one**** argument plus its personal illustration. Use transitions appropriate to the score level (e.g., "First," "Second," "Moreover" for higher scores; simpler connectors for lower scores).
3. ****Conclusion paragraph**** - restate thesis, summarize key points, and finish with a memorable closing or call-to-action.

7. Draft-revise-polish workflow (follow ****exactly****)

1. ****Parse the input**** - extract 'target_score', the chosen opening sentence, and its 'mean'/'std'.
2. ****Calculate bounds**** - compute 'lower' and 'upper'. Keep them in mind.
3. ****Plan**** - outline intro, each argument with personal example, and conclusion while considering the stylistic expectations for the score.
4. ****Draft**** - write full sentences, respecting the exact opening line (or adding the short bridge if needed). Include required arguments and examples.

```

5. **Count words** - split the draft on whitespace and count.
  - If the count is **outside** '[lower, upper]', **adjust organically**:
    * Add or expand a sentence that naturally deepens an argument, adds
      a relevant detail, or improves transition.
    * Remove or condense redundant wording only if it does not weaken
      required arguments or coherence.
  - Never insert filler text ("Lorem ipsum") or mechanically duplicate
    sentences.
6. **Polish** - verify:
  - Opening sentence is exact and first characters.
  - No prohibited word precedes it (bridge added only if necessary).
  - No title, quotation, or extra metadata is present.
  - Word count is within bounds.
  - Essay length and style match the target score.
  - All content requirements (position, >=2 arguments with personal
    examples, concluding paragraph) are met.
  - Sentence variety appropriate to the score, correct punctuation,
    capitalization, and overall fluency.
7. **Return** the final essay as plain text.

### 8. Common pitfalls to avoid
  - Adding any characters (even a space) before the required opening line.
  - Repeating the quotation from the prompt.
  - Exceeding the word-count window.
  - Omitting the required number of arguments or personal examples.
  - Using any of the prohibited first words **before** the opening sentence
    (the bridge, if needed, must also avoid them).
  - Including titles, headings, or any meta-information.
  - Making mechanical, non-organic changes to hit the word count.

By strictly adhering to the above procedure, the produced essay will
  satisfy every constraint and convincingly simulate a 10th-grade
  student's persuasive editorial. Good luck!"

```

A.4 GEPA Feedback Example

This is a snapshot of GEPA optimization process, which iteratively takes the previously generated output as examples and optimizes the essay generation instructions based on the feedback on linguistic feature alignments.

```

I provided an assistant with the following instructions to perform a task
  for me:
  ""
  <<Base prompt>>
  ""

```

The following are examples of different task inputs provided to the assistant along with the assistant's response for each of them, and some feedback on how the assistant's response could be better:

```

'''
# Example 1...5
## Inputs
<<Input used to generate the previous output>>

## Generated Outputs
<<Previously generated output>>

## Feedback
Evaluating the previously generated essay

Features out of the target distribution:
<<List of features>>
Features within the target distribution:
<<List of features>>

Do not mechanically repeat or remove content. Write your essay
organically to simulate the student's writing skills that corresponds
to the instructions.

'''
Your task is to write a new instruction for the assistant.

Read the inputs carefully and identify the input format and infer
detailed task description about the task I wish to solve with the
assistant.

Read all the assistant responses and the corresponding feedback. Identify
all niche and domain specific factual information about the task and
include it in the instruction, as a lot of it may not be available
to the assistant in the future. The assistant may have utilized a
generalizable strategy to solve the task, if so, include that in the
instruction as well.

Provide the new instructions within ''' blocks.

**Instruction for the "Essay-Simulation" Assistant**

'''
<<Revised instruction that follows the generation template>>

```

B Synthetic Essay Examples

The following examples show synthetic essays generated with different LLMs and prompt conditions for Essay Set 2 using GPT-OSS-20B.

```
# Essay Set #2
Type of essay: Persuasive/ Narrative/Expository
Grade level: 10
Average length of essays: 350 words
Rubric score range: 1-6
Final score range: 1-6

# Prompt
Censorship in the Libraries
"All of us can think of a book that we hope none of our children or any
other children have taken off the shelf. But if I have the right to
remove that book from the shelf -- that work I abhor -- then you also
have exactly the same right and so does everyone else. And then we
have no books left on the shelf for any of us." --Katherine Paterson,
Author
Write a persuasive essay to a newspaper reflecting your vies on
censorship in libraries. Do you believe that certain materials, such
as books, music, movies, magazines, etc., should be removed from the
shelves if they are found offensive? Support your position with
convincing arguments from your own experience, observations, and/or
reading.

# Rubric Guidelines
...
```

Base Prompt Score 1 Essay:

```
{
  "starting_sentences": {
    "A": "Apparently, the idea that books should be removed from
libraries seems unfair to me.",
    "B": "Clearly, teachers often think censorship keeps kids safe.",
    "C": "Possibly, my feelings about this topic are mixed.",
    "D": "Sometimes, people worry about the content in magazines.",
    "E": "Noticeably, some readers dislike seeing certain music in
schools."
  },
  "starting_sentence_pick": "C",
  "generated_essay": "Possibly, my feelings about this topic are mixed.
Clearly, many people think removing books is good. Also, many think
it restricts learning. Some schools remove books because of
complaints. However, these actions may hurt education. Parents often
ask for safer reading. Still, this limits choices. Some argue freedom
of books matters. To sum up, many think the debate continues."
}
```

Score 6 Essay:

```
{
  "starting_sentences": {
    "A": "Imagine a library where every book has been silenced by
anonymous hands.",
    "B": "Imagine a space where the shelves stand mute, each tome
locked behind unseen barriers.",
    "C": "Imagine walking into a hall where the musty silence
replaces the chatter of curious minds.",
    "D": "Imagine a sanctuary whose doors close on the stories many
are eager to read.",
    "E": "Imagine a world where every child\u2019s curiosity is
stifled by a curtain of censorship."
  },
  "starting_sentence_pick": "C",
  "generated_essay": "Imagine walking into a hall where the musty
silence replaces the chatter of curious minds. Such a scenario forces
us to weigh the value of free access against the comfort of
censorship. Because knowledge thrives on questioning, restricting
literature stifles critical thinking. Reflecting on my visits to the
campus library, I have witnessed students confront challenging titles
with curiosity, not fear. However, opponents argue that offensive
material may distort young minds and hinder community harmony.
Graceful as children\u2019s wonder may appear, their developmental
stage demands exposure to diverse perspectives rather than imposed
silence. Until now, school boards have rarely tasked
themselves with curating shelves to match parental whispers. Every
instance of removal\u2014whether it involves 'The Absolutely True
Diary of a Part-Time Indian' or a controversial documentary\u2014
fuels resentment toward authority. Most educators prioritize a
curriculum that equips students with analysis skills, not with
compliance. Furthermore, research indicates that exposure to complex
narratives cultivates empathy, essential for democratic discourse.
Moreover, when students discover that their books have been tampered
with, they become curious about the reasons behind such actions,
igniting investigative conversations. Consequently, transparency and
dialogue prove more effective than
blanket censorship in nurturing informed citizenship. Students who
were once skeptical of contentious stories began engrossing
themselves in discussions about morality, context, and authorship.
When curriculum designers wield filters, they risk creating a narrow
worldview, ultimately limiting the library\u2019s purpose as a
gateway to knowledge. Observing small debates in tutoring sessions, I
recognize that questioning rights produces mature arguments, not
passive agreement. Adding another dimension, scholars argue that
censorship often benefits a minority\u2019s agenda at the expense of
a society\u2019s collective growth. Paradoxically, the very books
removed often catalyze protests that
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broaden the community\u2019s engagement with the topic. Drawing from
real\u2011life cases such as the controversy over 'To Kill a
Mockingbird', these protests reinforced the value of preserving
literature, no matter the discomfort. Therefore, it follows that a
deliberate, reasoned approach that retains all texts while educating
users is preferable over arbitrary removal. Hence, libraries should
remain neutral custodians, preserving all voices rather than
selectively amplifying silence. Ultimately, the vitality of any
democratic society lies in confronting uncomfortable ideas, not in
fearing them. Let us empower our students to navigate literature
responsibly, equipping them with the tools to
discern, critique, and respect divergent viewpoints. So the call is
simple: keep the shelves full, confront the content, and\u2014after
all\u2014teach the art of critical thought."
}

```

Feature-Guided Prompt Score 1 Essay:

```

{
  "starting_sentences": {
    "A": "Seeing whether a library can hold every book is tricky.",
    "B": "Addressing censorship in libraries demands careful thought
    .",
    "C": "Debating whose books stay on the shelf feels intense.",
    "D": "Listening to people who want books removed can be hard.",
    "E": "Choosing what books are safe for readers worries many."
  },
  "starting_sentence_pick": "C",
  "generated_essay": "Debating whose books stay on the shelf feels
  intense. Some people say we should pull books that hurt feelings or
  seem wrong. They think removal stops bad influences. However, many
  other people, librarians included, believe every book matters. If we
  keep every book, people can see and think about new ideas. When the
  shelves are cleared, we also lose learning chances. For example, a
  student once wanted to read about a controversial topic for a school
  project, but the book was taken away. The student was upset that they
  could not learn. Even if some people think a book is poor, putting
  it somewhere else can let people decide for themselves. After this,
  keeping books in libraries helps to explore many different thoughts."
}

```

Score 6 Essay:

```

{
  "starting_sentences": {
    "A": "Curbing access to contested literature does little more
    than stifle intellectual growth.",
    "B": "When shelves empty themselves of daring titles, the very
    doors meant for learning swing shut.",

```

```

    "C": "Removing books that challenge our beliefs is more a
concession to fear than a guardian of values.",
    "D": "Silencing dissenting voices shortens discourse in a society
hungry for knowledge.",
    "E": "Allowing anonymous whims to dictate what students read
undermines the very purpose of libraries."
},
"starting_sentence_pick": "C",
"generated_essay": "Because librarians serve as guardians of
knowledge, they must resist pressure to remove books under vague
pretenses. Students who encounter diverse viewpoints acquire critical
thinking skills that cannot be cultivated in a homogenous collection
. Censorship undermines intellectual curiosity by presenting a
curated narrative that echoes only the dominant voice in the room.
Examples abound of titles that once sparked debate, later removed for
fear of backlash, and in doing so, libraries sacrificed their role
as forums for dialogue.\n\n
Legislative attempts to ban works often backfire, stoking public
interest and prompting activists to seek the very books they tried to
silence. Legislation that depends on subjective criteria\u2014
u201charmless to children\u201d and \u201cmorally suitable\u201d\u2014
u201dproduces a slippery slope where teachers and parents can censor
any material that challenges their worldview. Rather than stifle
learners, such measures forces them to seek alternatives outside
formal scholastic contexts, resulting in uneven and precarious access
to information.\n\n
Beyond politics, censorship strips readers of emotional resonance.
Dystopian novels remind us that tyranny thrives on misinformation,
while memoirs expose the lives of those marginalized by society.
Access to these stories cultivates empathy\u2014a quality essential
for democratic citizenship. Silencing these narratives deprives
students of the chance to confront uncomfortable truths, thereby
sanitizing history and denying them the ethical frameworks needed for
informed decision-making.\n\n
Alternatives to removal include contextualizing content. Librarians
can provide study guides, host discussions, and facilitate
conversations that illuminate potentially challenging sections.
Guided inquiry allows learners to frame objections in an academic
setting rather than through suppression. When educators employ
critical frameworks, students gain agency over their learning journey
while still respecting community standards. Cultivating such
practices encourages reflective stewardship rather than passive
acceptance.\n\n
Religious groups, parents, or political campaigns frequently champion
censorship as a defense of moral values. However, poor peer-reviewed
research suggests that restricting literature correlates with lower
educational achievement. Censorship does not protect children; it
limits curiosity and fortifies ignorance. Encountering controversial
viewpoints in a controlled environment teaches students how to

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evaluate arguments and discern credibility\u2014a lesson that reading
freedom uniquely delivers within libraries.\n\n
In the digital age, access to knowledge is no longer linear. The rise
of online databases and open-access journals presents a paradox:
while the internet democratizes information, it also amplifies
misinformation. In this context, librarians serve as navigators who
help readers filter, verify, and contextualize data. Removing books,
therefore, can paradoxically shrink the range of perspectives people
consult, rendering them ill-equipped to navigate a complex
information landscape.\n\n
Uniformly preserving unrestricted access strengthens democratic
ideals, cultivating leaders who can debate, negotiate, and uphold
pluralism. In communities where schools provide a non-judgmental
space for intellectual exploration, students are empowered to
question norms, thereby driving progress. When libraries become
repositories that respect the spectrum of ideas, they become
catalysts for societal resilience. Failure to protect these spaces
risks eroding the foundations upon which open discourse and
collective learning stand.\n\n
Ultimately, libraries should embrace their duty to oppose censorship,
ensuring that every reader\u2014regardless of background\u2014can
explore the breadth of human thought. Without such commitment, the
promise of libraries as equalizers of knowledge dissolves into
selective narrative control, compromising the very essence of
enlightenment."
}

```

Optimized Score 1 Essay:

```

{
  "starting_sentences": {
    "A": "The removal of controversial books from public libraries
restricts learning.",
    "B": "Books contain ideas that should be freely available to all
readers.",
    "C": "In the library, children can discover stories that
challenge their thinking.",
    "D": "Access to diverse literature is essential for developing
critical thinking.",
    "E": "When a book is banned, a community loses a voice."
  },
  "starting_sentence_pick": "D",
  "generated_essay": "Access to diverse literature is essential for
developing critical thinking. I believe we should not remove books
from libraries even if some people find them offensive. When I was in
ninth grade, the school librarian took a novel that taught about
kindness and saved many students who felt alone. If that book had
stayed, those children could have learned positive ways to help
others. I also see an older friend who feels upset because a guide on

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    mental health was removed. Keeping such books allows everyone to
    find help. Let us defend our libraries and make sure good books stay
    on the shelves."
}

```

Score 6 Essay:

```

{
  "starting_sentences": {
    "A": "Consider the uneasy silence that swallows a library hall
    when a controversial book is removed.",
    "B": "When a shelf clears of text, a community loses more than
    words.",
    "C": "Imagine a space where older tomes stand alone after a
    censorship decree.",
    "D": "If a book goes missing, both the mind and the story suffer
    .",
    "E": "Silence is louder when a library denies access to
    challenging ideas."
  },
  "starting_sentence_pick": "A",
  "generated_essay": "Consider the uneasy silence that swallows a
  library hall when a controversial book is removed. In a quiet Tuesday
  morning at Jefferson High, the librarian\u2019s face was washed in
  the pale light of the reading room, and she whispered that the new
  book about adolescent sexuality had been taken down because it
  offended some parents. That moment crystallized one truth for me:
  censorship erodes not the content but the very spirit of inquiry that
  schools are meant to nurture.\n\n
  First, removing offensive material narrows the horizons of young
  minds. When I was in sophomore year, I saw a peer ask, \u201cWhy can\u2019t
  we just read what we want?\u201d We were denied a novel that
  dealt with cultural identity, simply because it contained a scene
  with a character reflecting a marginalized background. That denial
  wasn\u2019t a protection; it was a condemnation of diversity.
  Libraries should be marketplaces of ideas, and when any insurance
  policy removes a key item, the entire collection loses flavor. In the
  context of literature, censorship replaces nuanced portrayals with
  sanitized clich\u00e9s, making it less likely that a student will
  encounter a viewpoint that challenges bias.\n\n
  Consider the school\u2019s own handbook, which includes a policy that
  restricts materials deemed \u201cinappropriate.\u201d Yet the policy
  also explicitly encourages diversity and exposure to a broad
  curriculum. The contradiction is stark: an institution designed to
  cultivate critical thinking paradoxically purges the very content
  that challenges assumptions. When students see this dissonance, they
  learn less about open inquiry and more about obedience to authority.\n\n
  Second, censorship undermines the trust between students and
  educators. When the committee decided to pull a bilingual memoir I

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had proudly cataloged, many of my peers were surprised and disappointed. They began to think that the library, instead of offering an objective space, might have been swayed by political whims. This mistrust extended beyond that single book: students stopped asking to find books that were previously available. Trust is foundational in education; any act perceived as censorship erodes that glue.\n\n

Furthermore, a culture that normalizes censorship subtly signals to students that dissent is dangerous. In a recent school\u2011wide survey, a significant percentage of students reported feeling uncomfortable discussing controversial topics in class, citing fear that a teacher might uncover a book they had checked out. This climate of self\u2011censorship is precisely what legislators fears; without dialogue, societal blind spots remain unchallenged, leading to stagnation.\n\n

Third, the practice of censorship stifles future innovation. When libraries remove books with provocative content, they discard future scholarship. I recall a senior project where a group was researching gender roles in contemporary media. Libraries had removed all to\u2011discussed texts because of another parent complaint, forcing the research group to use secondary summaries that lacked depth. Innovation thrives on questioning, and when questioning is precluded by market decisions, the entire field stalls. Libraries must be the crucible where the youth\u2019s curiosity can be tested without fear of reprisal.\n\n

The legal foundation that supports a free library is strong. The First Amendment protects not only the public\u2019s right to information but also the library\u2019s duty to provide diverse materials. When a school board removes a text, it risks legal challenges and sets a precedent for future restrictions that could erode the overall free\u2011speech climate in education.\n\n

Above all, the tenet of freedom of speech dictates that readers, not administrators, decide what is enlightening. When parents and community members demand that a particular book be removed, it is not an act of protecting children, but a threat to the library\u2019s primary mission. By decanting the dangerous content for safety, libraries protect an illusion of safety while committing an injury to education. The world does not have to be simplified because some stakeholders fear growth; instead, the library should offer multiple verifiable perspectives to teach critical thinking.\n\n

In conclusion, the decision to purge offensive material from library shelves ends up hurting the intellectual growth of the very population it seeks to protect. Censorship narrows worlds, breaks trust, blocks innovation, and betrays the sacred function of knowledge. Instead of stripping books, we should equip young scholars with the analytical tools to sift truth from rumor, to differentiate bias from fact, and to challenge their own preconceptions. I urge our community to reinstate the books at issue, reaffirm the library\u2019s role as a haven of diverse ideas, and protect the future by allowing a mind at liberty to explore, question, and grow."

}

C Selecting Handcrafted Features

C.1 Feature Importance Analysis

Table C1. Inclusion of each handcrafted feature in the top-5 list used for prompt optimization in each ASAP essay set (✓ indicates inclusion).

Feature	1	2	3	4	5	6	7	8
(Counts) Words appearing once (<code>unique_word</code>)	✓	✓	✓		✓	✓	✓	✓
(Counts) Essay length in characters (<code>ess_char_len</code>)	✓	✓	✓	✓	✓	✓	✓	
(Counts) Essay length in words (<code>word_count</code>)	✓	✓	✓	✓	✓	✓	✓	
(Proportion) Prepositions and commas (<code>prep_comma</code>)	✓	✓						✓
(Proportion) Past-tense verbs (<code>VBD</code>)				✓	✓		✓	
(Proportion) Present-tense verbs (<code>VBP</code>)			✓				✓	
(Proportion) Commas (,)	✓							
(Proportion) Base-form verbs (<code>VB</code>)			✓					
(Proportion) Possessive pronouns (<code>PRP\$</code>)				✓				
(Proportion) Adjectives proportion (<code>JJ</code>)						✓		
(Avg.) Positive sentiment words (<code>overall_positivity_score</code>)					✓			
(Avg.) Negative sentiment words (<code>overall_negativity_score</code>)						✓		
(Avg.) Word length in characters (<code>mean_word</code>)								✓
(Variance) Word-length (<code>word_var</code>)		✓		✓				✓
(Variance) Sentence-length in words (<code>sent_var</code>)								✓

As a preliminary step, we analyzed handcrafted feature importance for AES prediction using random forest models. The top-5 features varied by essay set but commonly included unique words, essay length (in words or characters), and part-of-speech proportions (Table C1).

C.2 LLM Capability to Follow Instructions with Handcrafted Feature Distributions

We conducted a preliminary evaluation to assess which handcrafted features provide significant alignment improvements when used in the LLM instructions. Using the candidate features from Table C1, we compared the alignment scores of synthetic essays generated under the Base and Feature-Guided conditions. The results showed that, across all LLMs, essay-length features (e.g., `word_count`) yielded significantly better alignment with authentic essay distributions under the Feature-Guided prompt condition.

Table C2. Average Gaussian similarity scores across features for each LLM (GM: Gemma3-12B, MN: Ministral3-14B, GO: GPT-OSS-20B) under the Base / Feature-Guided conditions. * indicates significant improvements over the Base condition (paired t-test $p < 0.05$). Higher values indicate better alignment.

Feature	GM	MN	GO
,	0.31 / 0.29	0.41 / 0.46	0.33 / 0.38
JJ	0.46 / 0.45	0.53 / 0.54	0.47 / 0.45
PRP\$	0.74 / 0.74	0.66 / 0.69	0.66 / 0.65
VB	0.52 / 0.52	0.55 / 0.53	0.55 / 0.53
VBD	0.72 / 0.71	0.66 / 0.69	0.67 / 0.66
VBP	0.63 / 0.62	0.71 / 0.71	0.66 / 0.61
ess_char_len	0.35 / 0.35	0.35 / 0.35	0.51 / 0.59*
mean_word	0.17 / 0.19	0.39 / 0.41	0.17 / 0.15
overall_negativity_score	0.68 / 0.68	0.67 / 0.66	0.66 / 0.68
overall_positivity_score	0.71 / 0.71	0.64 / 0.66	0.73 / 0.70
prep_comma	0.43 / 0.43	0.39 / 0.38	0.51 / 0.46
sent_var	0.80 / 0.82	0.81 / 0.82	0.80 / 0.78
unique_word	0.21 / 0.22	0.19 / 0.19	0.19 / 0.15
word_count	0.37 / 0.44*	0.30 / 0.45*	0.55 / 0.68*
word_var	0.20 / 0.18	0.33 / 0.34	0.30 / 0.21

D Checking Quality of Generated Essays with Handcrafted Feature Distributions

After generating synthetic essays under different prompt conditions, we calculated Glass’ delta effect sizes to measure alignment between authentic and synthetic essays. For this alignment analysis, we used all features listed in Table C1. The essay-generation prompts used `word_count` and `top-5` features.

Table D1 shows that the Feature-Guided and Optimized prompt conditions provided better alignment than the Base condition across all LLMs. The results for `top-5` features were similar to `word_count` or marginally worse, indicating that optimizing for essay length might be sufficient to improve overall alignment of handcrafted feature distributions.

E Augmented Training Set Results for All LLMs and Prompt Conditions

Table D1. Median Glass’ delta effect sizes measuring the overall alignment between authentic essays and synthetic essays generated under different prompt conditions and LLMs. Metrics were calculated across all score levels of each essay set. Lower values indicate better alignment (GM: Gemma3-12B, MN: Ministral3-14B, GO: GPT-OSS-20B).

Condition	LLM	Feature	1	2	3	4	5	6	7	8	Avg. ($\downarrow$)
Base	GM	N/A	0.573	0.671	1.019	1.489	0.735	0.642	0.595	0.689	0.802
	MN		0.697	0.359	0.709	0.982	0.859	1.154	0.769	0.406	0.742
	GO		0.680	0.688	0.641	0.880	0.749	0.387	0.680	0.872	0.697
Feature-Guided	GM	word_count	0.648	0.707	1.016	1.526	0.650	0.411	0.658	0.640	0.782
		top-5	0.582	0.623	1.070	1.610	0.927	0.569	0.698	0.840	0.865
	MN	word_count	0.771	0.303	0.727	0.941	0.911	0.916	0.754	0.434	0.719
		top-5	0.780	0.311	0.756	0.945	0.885	1.076	0.770	0.450	0.747
	GO	word_count	0.644	0.718	0.642	0.711	0.752	0.421	0.498	0.906	0.662
		top-5	0.568	0.736	0.637	0.755	0.804	0.382	0.502	0.925	0.664
Optimized	GM	word_count	0.584	0.742	0.657	1.171	0.442	0.219	0.421	0.576	0.602
		top-5	0.554	0.613	1.174	1.173	0.263	0.386	0.565	0.877	0.701
	MN	word_count	0.621	0.308	0.713	0.842	0.467	0.397	0.243	0.391	0.498
		top-5	0.705	0.296	0.661	0.915	0.398	0.473	0.453	0.357	0.532
	GO	word_count	0.309	0.884	0.509	0.525	0.530	0.228	0.319	0.441	0.468
		top-5	0.474	0.610	0.433	0.448	0.373	0.430	0.325	0.628	0.465

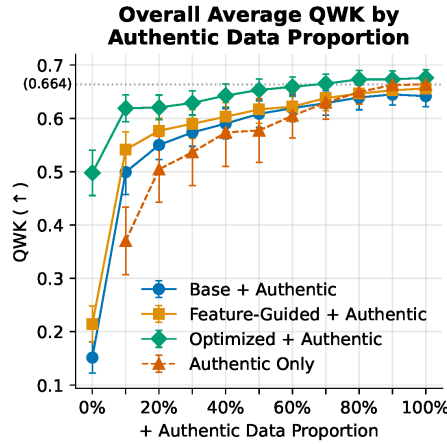


Fig. E1. Average QWK performance across essay sets for AES models with augmented data. Dotted lines indicate the authentic-only baselines with corresponding data proportions, along with horizontal lines for the 100% authentic data performance. Higher values indicate better performance.

Table E1. Average QWK scores of AES models trained on a mix of synthetic essays (from different LLMs and prompt conditions) and varying proportions of authentic essays. Higher values indicate better performance.

Data Source	Auth. Ratio	1	2	3	4	5	6	7	8	Avg.(↑)
Authentic only	0.1	0.402	0.489	0.500	0.550	0.547	0.386	0.089	0.000	0.370
	0.2	0.565	0.594	0.554	0.619	0.669	0.533	0.496	0.006	0.504
	0.3	0.625	0.612	0.572	0.620	0.691	0.587	0.561	0.024	0.537
	0.4	0.666	0.640	0.584	0.650	0.713	0.636	0.652	0.047	0.574
	0.5	0.656	0.633	0.570	0.654	0.712	0.635	0.670	0.088	0.577
	0.6	0.660	0.633	0.576	0.645	0.714	0.657	0.675	0.281	0.605
	0.7	0.666	0.648	0.587	0.659	0.723	0.660	0.692	0.401	0.629
	0.8	0.679	0.650	0.585	0.659	0.734	0.681	0.687	0.519	0.649
	0.9	0.667	0.640	0.594	0.682	0.733	0.692	0.695	0.599	0.663
	1.0	0.680	0.647	0.600	0.670	0.721	0.701	0.699	0.590	0.664
Base	0.1	0.550	0.534	0.444	0.576	0.658	0.559	0.463	0.210	0.499
	0.2	0.573	0.593	0.509	0.581	0.670	0.584	0.497	0.394	0.550
	0.3	0.599	0.594	0.524	0.615	0.696	0.607	0.535	0.420	0.574
	0.4	0.626	0.611	0.535	0.618	0.700	0.645	0.576	0.410	0.590
	0.5	0.640	0.616	0.539	0.633	0.713	0.652	0.615	0.461	0.609
	0.6	0.656	0.631	0.545	0.652	0.726	0.658	0.616	0.468	0.619
	0.7	0.658	0.635	0.565	0.658	0.718	0.673	0.626	0.497	0.629
	0.8	0.664	0.646	0.570	0.663	0.732	0.683	0.637	0.520	0.639
	0.9	0.670	0.649	0.568	0.674	0.726	0.686	0.630	0.556	0.645
	1.0	0.665	0.652	0.559	0.665	0.729	0.678	0.631	0.555	0.642
Feature-Guided	0.1	0.618	0.554	0.458	0.576	0.659	0.587	0.550	0.327	0.541
	0.2	0.623	0.597	0.511	0.586	0.685	0.610	0.566	0.438	0.577
	0.3	0.638	0.596	0.526	0.625	0.699	0.626	0.592	0.417	0.590
	0.4	0.659	0.610	0.544	0.603	0.707	0.650	0.616	0.437	0.603
	0.5	0.657	0.613	0.567	0.620	0.722	0.671	0.643	0.442	0.617
	0.6	0.674	0.635	0.538	0.637	0.719	0.675	0.641	0.458	0.622
	0.7	0.682	0.640	0.573	0.656	0.727	0.684	0.645	0.499	0.638
	0.8	0.682	0.635	0.570	0.670	0.730	0.686	0.660	0.542	0.647
	0.9	0.676	0.661	0.572	0.668	0.722	0.685	0.661	0.576	0.653
	1.0	0.678	0.645	0.568	0.668	0.732	0.701	0.668	0.591	0.656
Optimized	0.1	0.667	0.617	0.560	0.650	0.730	0.635	0.614	0.477	0.619
	0.2	0.645	0.642	0.582	0.641	0.720	0.645	0.613	0.478	0.621
	0.3	0.662	0.632	0.588	0.646	0.731	0.647	0.636	0.490	0.629
	0.4	0.677	0.651	0.590	0.662	0.734	0.667	0.650	0.511	0.643
	0.5	0.679	0.644	0.592	0.668	0.747	0.684	0.667	0.545	0.653
	0.6	0.681	0.657	0.583	0.676	0.744	0.684	0.671	0.578	0.659
	0.7	0.684	0.658	0.598	0.673	0.748	0.698	0.672	0.590	0.665
	0.8	0.691	0.667	0.602	0.694	0.743	0.696	0.683	0.610	0.673
	0.9	0.685	0.658	0.599	0.689	0.746	0.688	0.686	0.633	0.673
	1.0	0.692	0.673	0.596	0.691	0.742	0.699	0.677	0.640	0.676